

Souvik Pramanik

✉ pramanik-souvik@protonmail.com — 📞 +880 17 1817 0955
🌐 [pramaniksouvik](#) — 📄 [pramanik-souvik](#) — 🌐 [Portfolio](#)

Profile

Research Assistant at North South University and AI Researcher specializing in Natural Language Processing (NLP), Large Language Models (LLMs), Multimodal Learning, and Deep Learning. Author of 11+ research publications and preprints spanning Bengali NLP, sentiment analysis, multimodal AI, computer vision, healthcare AI, and explainable machine learning. Experienced in transformer fine-tuning, benchmark development, dataset engineering, and reproducible AI research, with a strong interest in low-resource language technologies and multilingual intelligent systems.

Education

North South University

2021 – 2025

Bachelor of Science in Computer Science and Engineering

Relevant Coursework: Data Structures & Algorithms, Machine Learning, Deep Learning, Natural Language Processing, Artificial Intelligence, Database Systems, Software Engineering, Operating Systems, Statistics, Theory of Computation, Discrete Mathematics.

Research Experience

Research Assistant

2025 – Present

North South University, Department of Computer Science and Engineering

- Contribute to research projects in Natural Language Processing (NLP), Large Language Models (LLMs), Multimodal Learning, and Machine Learning.
- Assist in dataset collection, preprocessing, annotation, benchmarking, and evaluation of deep learning models for research applications.
- Develop and optimize transformer-based architectures including BERT, RoBERTa, DeBERTa, BanglaBERT, XLM-RoBERTa, and LLaMA-based models.
- Conduct experiments, statistical analyses, and performance evaluations across multilingual and low-resource language datasets.
- Collaborate with faculty members and researchers on manuscript preparation, experimental design, literature review, and publication activities.
- Build reproducible machine learning pipelines using PyTorch, Hugging Face Transformers, PEFT/LoRA, and automated data-processing workflows.

Independent AI Researcher

2023 – Present

- Published and co-authored 11+ research papers and preprints in NLP, LLMs, Multimodal Learning, Computer Vision, Healthcare AI, and Explainable AI.
- Developed Bengali NLP resources, sentiment analysis systems, multimodal benchmarks, and low-resource language datasets.
- Designed automated data collection, preprocessing, annotation, and evaluation pipelines for multilingual machine learning research.

Research Interests

Natural Language Processing (NLP), Large Language Models (LLMs), Multimodal Learning, Low-Resource Language Technologies, Computational Linguistics, Retrieval-Augmented Generation (RAG), Explainable AI, Benchmark Development, Representation Learning, AI for Social Good.

Selected Publications

First Author

- **MultiSoc-4D: A Benchmark for Diagnosing Instruction-Induced Label Collapse in Closed-Set LLM Annotation of Bengali Social Media** (Preprint, 2026)

- **Bengali MemeNet (BMN): A Multimodal Framework for Sentiment Classification of Bengali Memes** (Preprint, 2026)

Co-Author Publications

- Enhancing BiGRU with a KAN Block for Legal Document Classification and Summarization (Preprint, 2026)
- Advancing Sentiment Analysis with BERT: A Comparative Evaluation Against Machine Learning and Deep Learning Techniques (IEEE SATC 2026)
- Balancing Performance and Sustainability in Vision-Language Models for Hateful Meme Detection with Code-Carbon Emission Analysis (Preprint, 2026)
- Bridging the Credit Gap: A Hybrid Decentralized Framework for Thin-File Borrowers using Stacking Ensembles and Blockchain Oracles (Preprint, 2026)
- Feature Engineering and Ensemble Classifiers for Robust Cardiovascular Disease Detection (ICSADL 2026)
- Leveraging Vision-Language Models for Movie Genre Prediction: A Comparative Analysis of BLIP, CLIP, ViLT, and FLAVa (IEEE SATC 2026)
- Beyond Classical Limitations: Improving Bone Fracture Classification in Data-Constrained Regimes via Quantum-Classical Synergy (Preprint, 2025)

Technical Skills

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| • Programming: Python, C, C++, Java, JavaScript, Bash | • Computer Vision: OpenCV, Vision Transformers, CLIP, BLIP |
| • Deep Learning: PyTorch, TensorFlow, Keras | • Data Science: Pandas, NumPy, Matplotlib, Seaborn |
| • Machine Learning: Scikit-Learn, XGBoost, LightGBM | • Data Collection: BeautifulSoup, Selenium |
| • NLP/LLMs: Hugging Face Transformers, PEFT, LoRA, RAG, LangChain, Tokenizers, Embeddings | • Tools: Git, Linux, Docker, VS Code, Jupyter Notebook |
| | • Experiment Tracking: TensorBoard, Weights & Biases |
| | • Databases: MySQL, SQLite |

Selected Projects

Bengali Sentiment Analysis on Natok and Review Datasets

- Conducted comprehensive sentiment classification experiments on Bengali drama and review datasets using traditional machine learning and transformer-based models.
- Evaluated BanglaBERT, BERT, RoBERTa, DeBERTa, XLNet, ALBERT, and DistilBERT architectures.
- Achieved strong classification performance through data preprocessing, augmentation, and systematic hyperparameter optimization.

Multimodal Hateful Meme Detection using Vision-Language Models

- Developed multimodal classification pipelines combining visual and textual information for hateful meme detection.
- Evaluated multiple fusion strategies and vision-language architectures for robust content moderation tasks.
- Performed carbon-emission tracking and sustainability analysis using CodeCarbon during model training and evaluation.

Multi-Threaded HTTP Server in Modern C++

- Developed a lightweight HTTP server from scratch using modern C++17.
- Implemented request handling, static file serving, concurrency management, and thread-pool based execution.
- Applied systems programming concepts including socket programming, networking, and concurrent processing.

Additional Information

Languages: Bengali (Native), English (Professional Proficiency)

Professional Interests: Large Language Models, Multimodal AI, Computational Linguistics, Dataset Engineering, Benchmark Development, Reproducible Machine Learning, Research Automation.